

MD MORSHED JAMAN

COMPUTER SCIENCE AND ENGINEERING GRADUATE | SOFTWARE DEVELOPMENT | DATA AND AI RESEARCH | SOFTWARE QUALITY

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[LinkedIn](#) | [GitHub](#) | [Live Portfolio](#) | [Portfolio GitHub](#)

[LeetCode](#) | [Codeforces](#) | [GeeksforGeeks](#)

Professional Summary

Computer Science and Engineering graduate from BRAC University with a CGPA of 3.71/4.00 and multidisciplinary experience in full-stack web development, databases, REST APIs, software testing, data handling, cloud-native workflows, cybersecurity, IoT systems, computer graphics, and healthcare AI research. Built independent and academic projects using React, Node.js, Express.js, Flask, SQL and NoSQL databases, Docker, GitHub Actions, authentication, dashboards, and responsive interfaces. Experienced in research, technical documentation, troubleshooting, customer communication, tutoring, and project coordination. Solved 500+ programming problems and currently strengthening data-science and analytical skills through professional training.

Education

BRAC University — Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

CGPA: 3.71/4.00 | Completed all degree requirements; awaiting official certificate.

Relevant Areas: Data Structures, Algorithms, Database Systems, Software Engineering, Web Development, Artificial Intelligence, Computer Graphics, Operating Systems, Computer Networks, Cybersecurity, Data Engineering, Project Management, and IoT Security.

Cantonment Public School and College, Saidpur

Saidpur, Bangladesh

Higher Secondary Certificate, 2019 | Secondary School Certificate, 2017

Technical and Professional Skills

Programming: Python, JavaScript, TypeScript Basics, C, C++, Java Fundamentals, PHP, SQL, Assembly, Verilog

Frontend: React.js, Vite, HTML5, CSS3, Tailwind CSS, Bootstrap, React Router, Context API, Axios, Responsive Design

Backend and APIs: Node.js, Express.js, Flask, Laravel Basics, REST APIs, JSON, JWT Authentication, Role-Based Access Control, CRUD, Input Validation, Error Handling

Databases and Data: MySQL, PostgreSQL Basics, MongoDB, SQLite, Redis, Mongoose, SQLAlchemy, SQL Joins, Filtering, Aggregation Concepts, Data Validation, Spreadsheet Reporting

Testing and Quality: Manual Testing, Functional Testing, Smoke Testing, Regression Testing, Test Cases, Bug Reporting, Postman, Selenium Fundamentals, Jira Concepts, Chrome DevTools, SDLC, STLC, Agile/Scrum

Cloud and Engineering Tools: Git, GitHub, Docker, Docker Compose, GitHub Actions, Kubernetes Basics, Helm Basics, Linux, VS Code, Figma, Canva, Microsoft Office, Google Workspace

Research and Emerging Technology: Machine Learning Fundamentals, Computer Vision, Medical Image Analysis, Super-Resolution, Foundation-Model Adaptation, Uncertainty Estimation, YOLO Exposure, MQTT, ESP32, Cryptography Concepts, OpenGL, OpenAI API Integration

BI and Analytics Exposure: Microsoft Excel, Google Sheets, Operational Reporting, Data Cleaning Basics, Charts, KPI-Oriented Analysis, Power BI Fundamentals, Tableau Fundamentals

Research Experience

Lightweight Hybrid Security Framework for MQTT-Based Medical IoT Systems

Completed Undergraduate Thesis

- Designed and evaluated an ESP32-to-MQTT medical-sensor workflow using Python, JSON, encryption, pseudonymization, authentication tags, replay prevention, tamper detection, and authorized gateway decryption.
- Conducted attack simulations and analyzed latency, throughput, protocol overhead, scalability, reliability, and deployment readiness; documented the findings in a 92-page thesis.

Quality-Aware Fetal Ultrasound Enhancement (QAFUE)

Completed / Unpublished

- Developed a perceptual super-resolution and acquisition-aware feedback framework for fetal-ultrasound enhancement and validated it using the HC18 and ACOUSLIC-A datasets.
- Worked on dataset preparation, image-quality assessment, anatomy preservation, experimental comparison, result visualization, and manuscript preparation.

Cross-Device Fetal Ultrasound Adaptation

Ongoing

- Investigating uncertainty-guided adaptation of fetal-ultrasound foundation models for reliable performance across devices, domains, and acquisition conditions.

Tau-PET Progression Forecasting

Ongoing

- Exploring multimodal and temporal methods for forecasting regional tau-PET progression using imaging, cognitive, clinical, and biomarker data with uncertainty and missing-modality analysis.

Experience and Leadership

Founder and Directing Manager, Online Clothing Page

Facebook Page

- Manage product listings, customer communication, order coordination, digital content, sales support, and routine operational records.
- Use customer feedback and product-interest patterns to improve product presentation, communication, and daily digital operations.

Private Tutor and Local Coaching Centre Instructor

4 Years / Semester Breaks

- Provided private tuition for four years and taught at a local coaching centre during university semester breaks, supporting students with lessons, homework, examination preparation, and difficult concepts.

Selected Software and Data Projects

Cloud-Native Products Platform

[GitHub](#)

React, TypeScript, Node.js, Express.js, PostgreSQL, Redis, Docker, GitHub Actions

- Built a product-management application with REST APIs, PostgreSQL persistence, Redis caching, CRUD workflows, validation, structured error responses, health checks, database migrations, and a responsive frontend.
- Containerized services and practiced CI/CD, backend testing, cache invalidation, environment configuration, service integration, and deployment debugging.

TaskFlow Task Manager

[GitHub](#)

React, Node.js, Express.js, MongoDB, JWT, Tailwind CSS

- Developed a multi-role task-management system with registration, authentication, protected routes, task CRUD, search, filtering, sorting, status tracking, dashboard analytics, and administrator workflows.
- Implemented validation, permissions, reusable components, API integration, error handling, and responsive user experiences.

NutriSnap Nutrition Platform

[GitHub](#) | [Live Demo](#) | [Figma](#)

React, Flask, MySQL/SQLite, SQLAlchemy, Bootstrap, Chart.js, APIs

- Built a multi-role nutrition platform with meal logging, vendor discovery, nutrition tracking, dashboards, search, reviews, water tracking, moderation, and AI-generated nutrition explanations.
- Worked with authentication, relational data, external APIs, role-specific workflows, Figma prototypes, forms, charts, and database-backed reporting.

CryptoBlog Web

[GitHub](#)

JavaScript, Node.js, Express.js, MySQL

- Developed a security-focused content platform with authentication, role-based access, protected routes, password hashing, database-backed content management, and secure workflow concepts.

Personal Portfolio

[GitHub](#) | [Live Website](#)

React, Responsive Frontend Development

- Created a personal portfolio to present software projects, research, technical skills, certifications, achievements, teaching experience, entrepreneurship, and professional profiles.

Additional Technical Projects

Maze3D Game: Python and OpenGL project involving 3D scene construction, camera control, coordinate transformations, movement, interaction, and game-state logic.

Event Management System: Laravel, PHP, MySQL, forms, routing, MVC concepts, and database-backed event-management workflows.

Hostel Meal Monitoring System: Google Sheets system for member records, meal counts, deposits, expenses, balances, monthly calculations, data validation, and operational reporting.

Firefighting Robot and Smart Safety Helmet: Hands-on hardware projects involving sensor integration, safety monitoring, embedded-system workflows, and YOLO-based image-processing exposure.

WordPress UI Projects: [Gizmo](#) and [Sequence](#); responsive layouts, navigation, typography, content organization, and visual presentation.

Certifications and Training

Data Science for ITEE — Semiconductor and ICT Training

Ongoing

[Data Collection and Processing with Python](#) — University of Michigan, Coursera

[Introduction to Data Engineering](#) — IBM, Coursera

[Foundations of User Experience \(UX\) Design](#) — Google, Coursera

[Foundations of Project Management](#) — Google, Coursera

[Project Initiation: Starting a Successful Project](#) — Google, Coursera

[Project Planning: Putting It All Together](#) — Google, Coursera

[Introduction to Software Testing](#) — University of Minnesota, Coursera

[Introduction to Selenium](#) — Coursera

Achievements, Activities and Languages

- Solved **500+ programming problems** across LeetCode, Codeforces, GeeksforGeeks, and other platforms, strengthening algorithms, debugging, pattern recognition, and edge-case analysis.
- Received a merit-based scholarship at BRAC University for academic performance.
- Member of the BRAC University Computer Club and Event Management Club; participated in workshops, volunteering, academic events, and team-based activities.
- **Languages:** Bangla — Native; English — Professional working proficiency.
- **Strengths:** Analytical thinking, attention to detail, ethical awareness, teamwork, adaptability, ownership, communication, documentation, and willingness to learn.